Analyze the sustainability of your storage systems

Digital Advisor

NetApp
March 28, 2024

This PDF was generated from https://docs.netapp.com/us-en/active-iq/learn_BlueXP_sustainability.html on March 28, 2024. Always check docs.netapp.com for the latest.

Table of Contents

Analyze the sustainability of your storage systems . . . 1
- Learn about sustainability . . . 1
- Access Sustainability dashboard . . . 2
- Sustainability dashboard overview . . . 4
- Improve sustainability score . . . 6

Analyze the sustainability of your storage systems

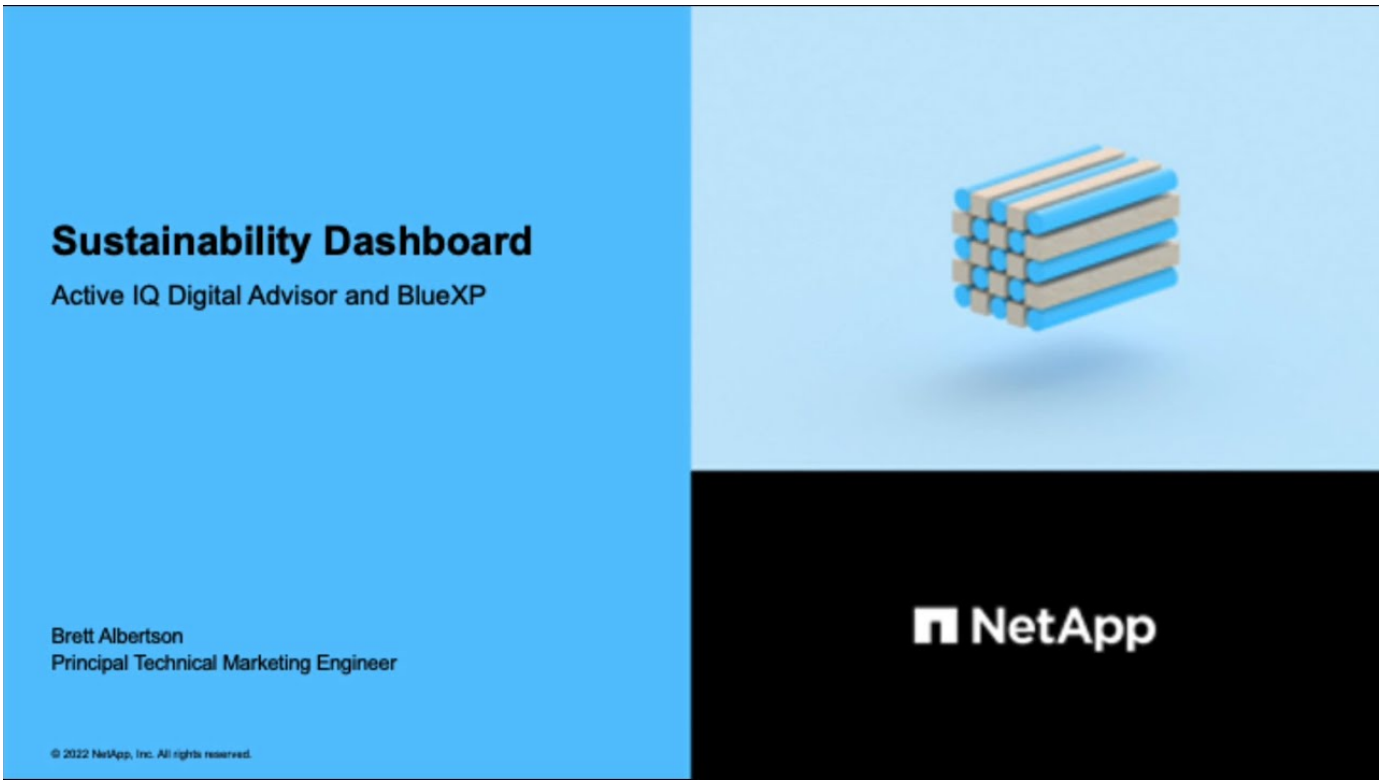
Learn about sustainability

Sustainability, as a service, allows you to reduce the energy consumption and enables you to work towards your environmental goals. This enables you to align your storage systems with eco-friendly practices and achieve the sustainability targets.

You can use the Sustainability dashboard to view the sustainability score, and projected usage of power, direct carbon, and heat from the sustainability dashboard. You can adjust the carbon mitigation percentage for specific sites. You can also view the sustainability score at the cluster level. Based on the sustainability score, you can assess the sustainability posture and implement NetApp's recommended actions to improve the score. To learn more about the Sustainability dashboard, go to Sustainability dashboard overview.

> Sustainability is supported on AFF systems, both A-Series and C-Series, FAS systems, and Cloud Volumes ONTAP.

You can view the following video to understand the Sustainability dashboard:



Benefits of sustainability

Sustainability offers the following benefits:

- Accelerate performance by reducing the number of storage devices to store the same amount of data.
- Lower storage costs by optimizing storage systems utilization.

- Reduce carbon footprints by using renewable energy at data centers.
- Improve energy efficiency by implementing energy-efficient policies.

Access Sustainability dashboard

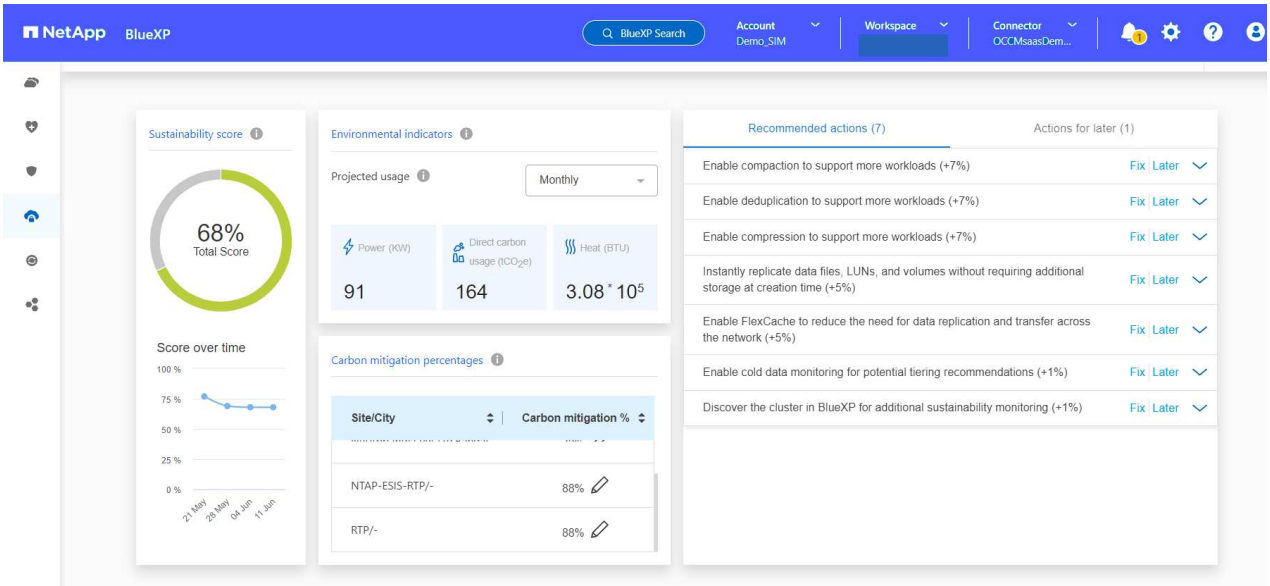
Sustainability dashboard provides AIOps-enabled reporting and scoring with actionable insights to improve your sustainability posture. You can access the Sustainability dashboard through NetApp BlueXP or Digital Advisor.

BlueXP

To log in to BlueXP, you can use your NetApp Support Site credentials or you can sign up for a NetApp cloud login using your email and a password. Learn more about logging in to BlueXP.

Steps

1. Open a web browser, and go to the BlueXP console.
   The NetApp BlueXP login page appears.
2. Log in to BlueXP.
3. From the BlueXP left navigation, select **Governance** > **Sustainability**.
   The Sustainability dashboard appears.



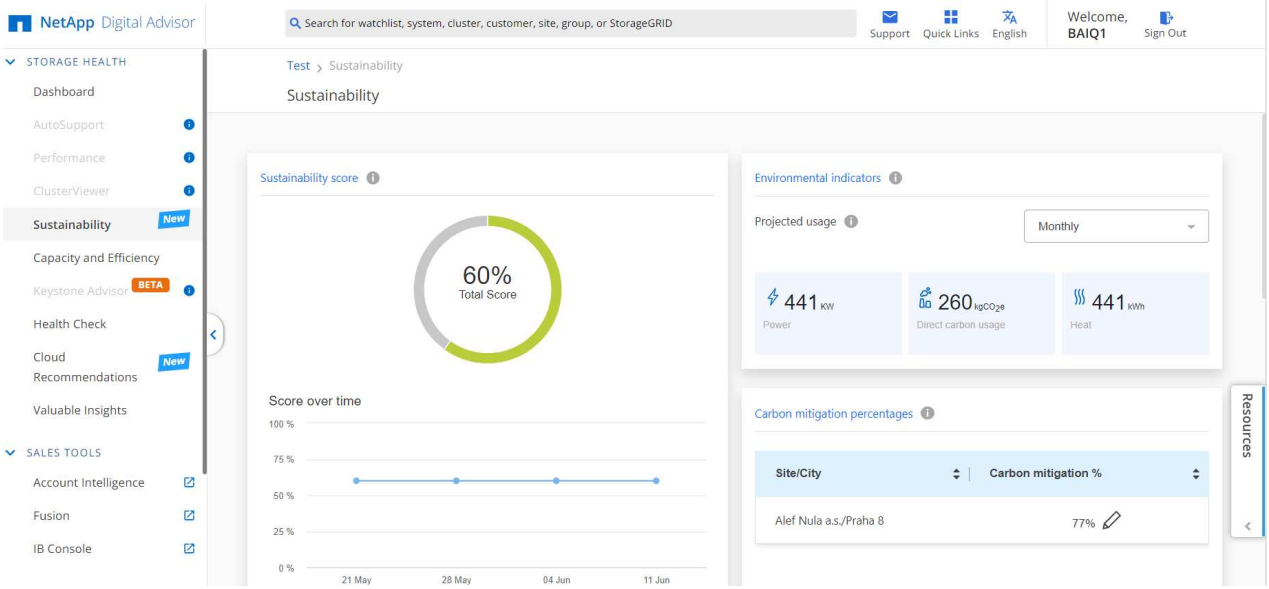
> If the Sustainability dashboard is not set up, the option **Add NSS account** appears. Provide your NetApp Support Site (NSS) credentials to view your Sustainability dashboard and the systems associated with your account.

Digital Advisor

To log in to Digital Advisor, you can use your NetApp Support Site credentials.

Steps

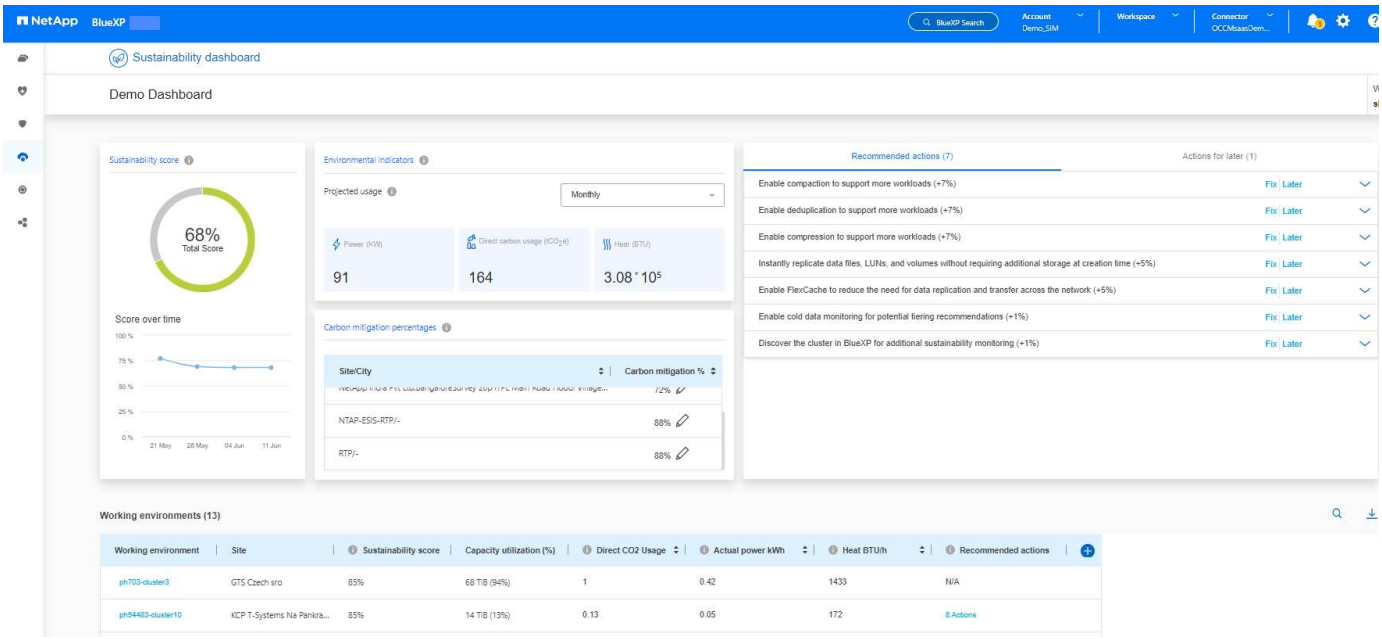
1. Open a web browser, and go to the Digital Advisor login page.
2. Provide your username and password and click **Sign In**.
3. From the left navigation, select **STORAGE HEALTH** > **Sustainability**.



The **Sustainability** option is disabled if the watchlist is not configured. To enable it, you can either create a watchlist or perform a search using customer name, site name, group name, StorageGRID, hostname, cluster, serial number, or system ID. Learn more about watchlist.

Sustainability dashboard overview

Sustainability dashboard provides an environmental assessment of your storage system and actionable insights for improvement in the form of NetApp's recommended actions.



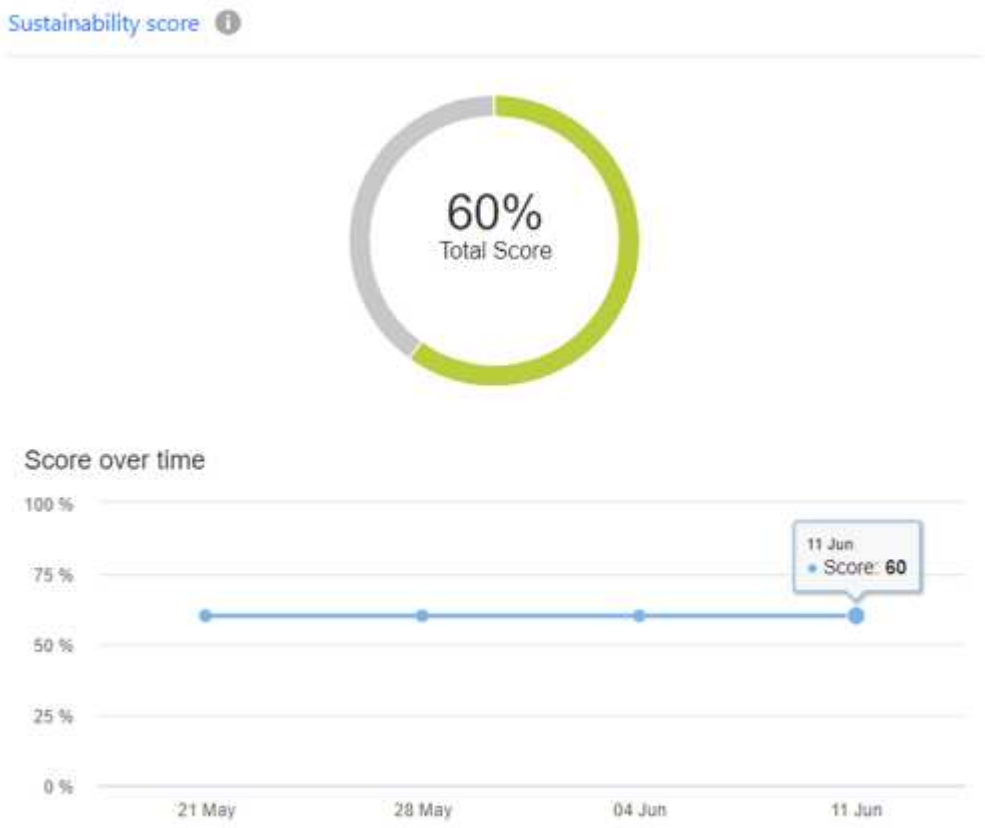
AutoSupport should be enabled for accurate calculation of the sustainability score.

Sustainability score: Displays the total score indicating the environmental sustainability of your storage

systems. You can assess the sustainability level of your storage systems based on the following range:

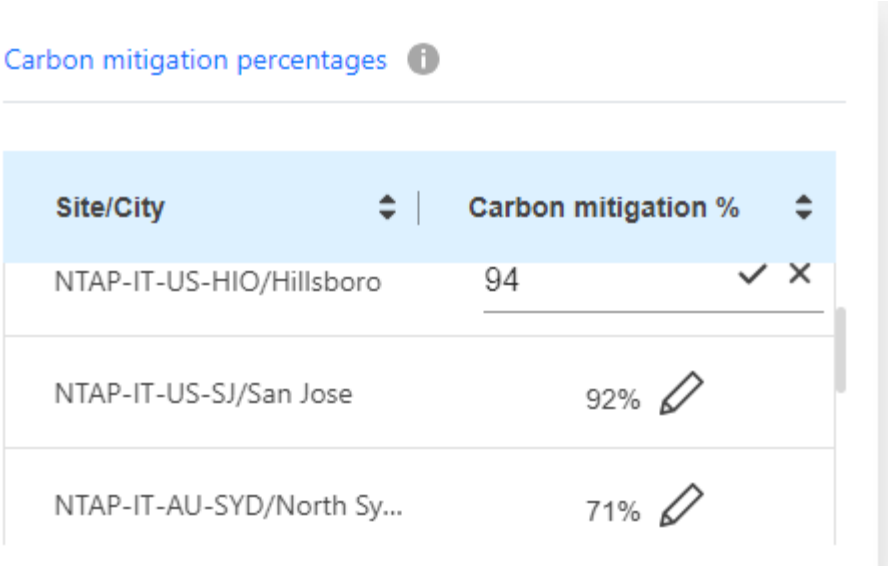
- 76 - 100: Indicates that sustainability is a top priority.
- 51 - 75: Indicates a high level of investment in sustainability initiatives.
- 26 - 50: Indicates good progress towards sustainability initiatives.
- Less than 25: Indicates the need for improvement in sustainability practices.

You can see the graphical representation of the score for up to 5 weeks, which is updated on a weekly basis. You can also see the reason for the increase or decrease in the sustainability score by hovering over the graph.



Environmental indicators: Displays projections for your power, direct carbon usage, and heat emission to evaluate the environmental health of your storage system. These projections are based on actual power, if unavailable, typical power values. You can view these projections monthly, quarterly, or yearly by selecting the desired option from the drop-down in the top right corner of this section.

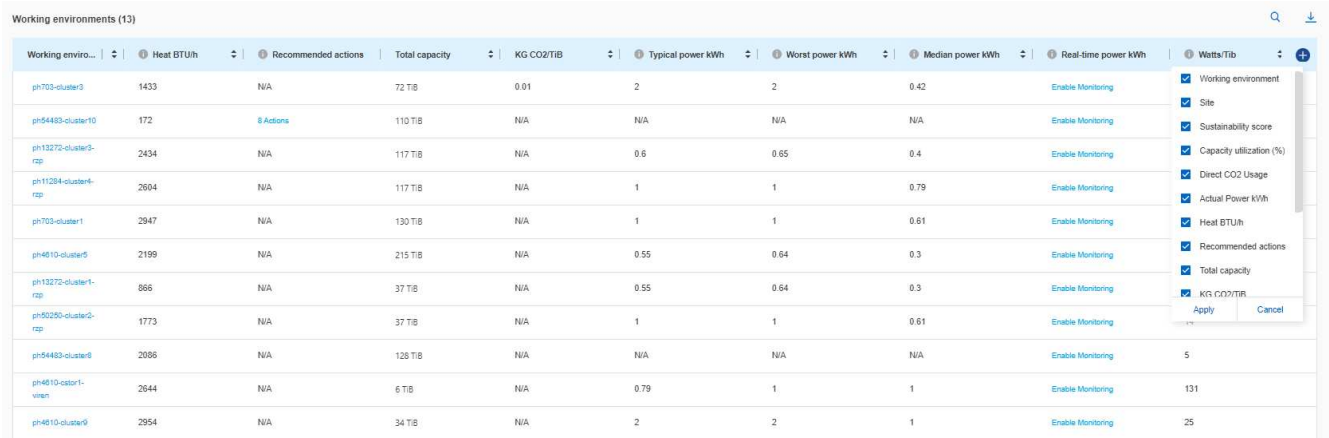
Carbon mitigation percentages: Displays the percentage of carbon mitigation at each site/city, and the presented baseline values are based on your location. You can adjust the carbon mitigation percentage for specific sites by clicking the icon located next to the percentage values, and the carbon numbers will automatically adjust accordingly.



Recommended actions: Displays the list of recommended actions to improve the sustainability score of your storage system. These actions can be taken immediately, or later.
For more information on how to improve the sustainability score, see Improve sustainability score.

Working environments: Displays environmental parameters in the table that can help in identifying clusters for moving to more efficient storage. From this table, you can:

- View the sustainability score at cluster level, select the cluster name to access **ClusterViewer**, take specific recommended actions to improve the sustainability. For more information, see Improve sustainability score at cluster level.



- Enable monitoring from the **Real-time power** column to view the real-time power details at the cluster level. For more information, see Cloud Insights.

Improve sustainability score

Sustainability dashboard provides recommended actions that you can implement to improve the overall sustainability score and the cluster-level sustainability score.

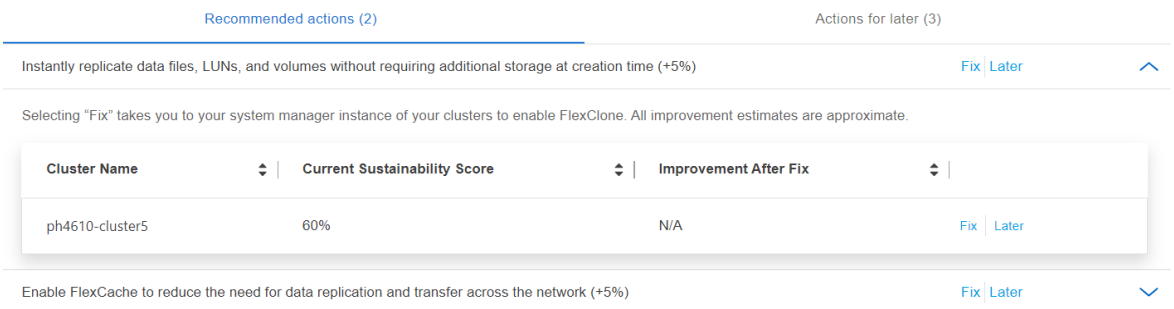
To get the best results from the dashboard, you must enable AutoSupport. If AutoSupport is not enabled, the data will be based on product specifications. To get real-time power details, you need to enable Cloud Insights. For more information on how to enable Cloud Insights, see Cloud Insights.

Improve overall sustainability score

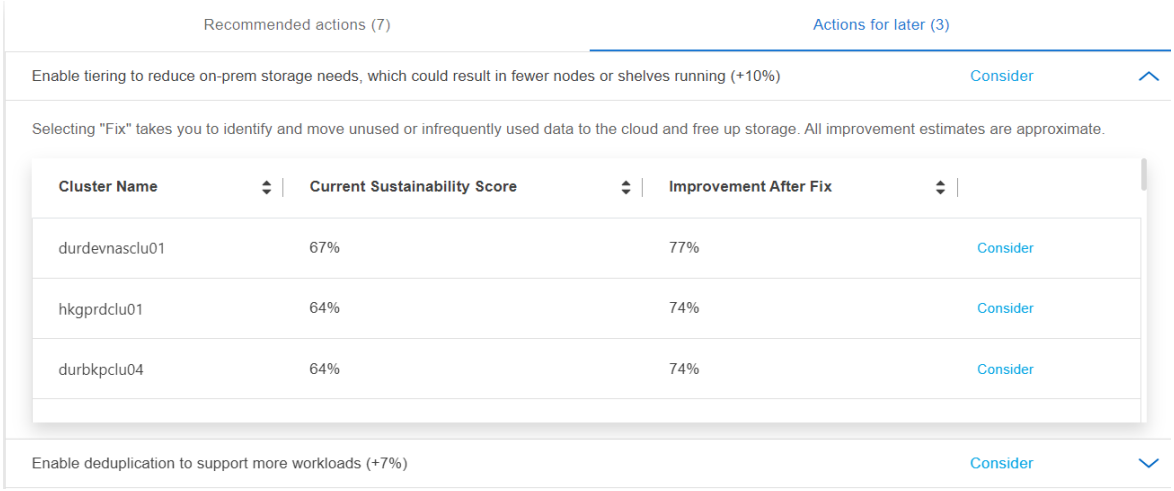
You can follow these steps to improve the overall sustainability score and that focuses on sustainability efforts at company level:

BlueXP

1. Go to **Sustainability** from the **Governance** category available in the BlueXP left navigation.
2. Go to the **Recommended actions** tab.
   - You can select **Fix** to take these actions immediately, or select **Later** to address these actions later.
   - If you plan to address the actions immediately, select **Fix**.
     - It expands the view of the selected recommended action. You can also expand the recommended action view using the down arrow. In the expanded view, you can see the cluster name, the sustainability score, and subsequent increase if you proceed with the **Fix** option.



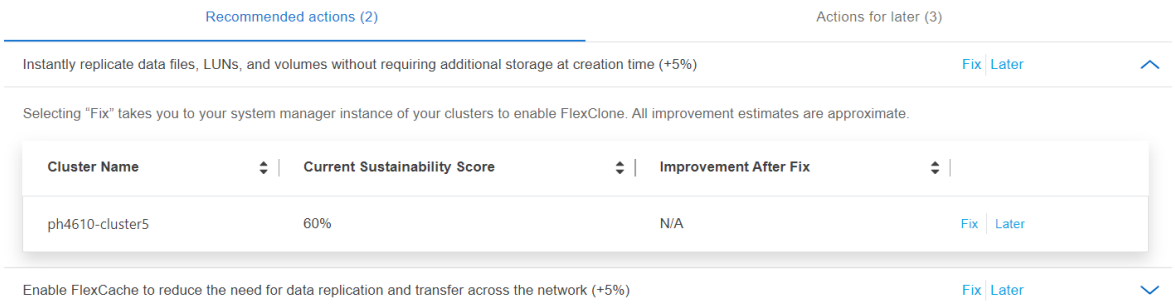
   - If you plan to address the actions later, select **Later**.
     - If you select **Later**, it moves the selected recommended action to the **Actions for later** tab. The selected action will be postponed for 30 days. After 30 days, this action will move to the **Recommended actions** tab.
     - You can also review the postponed actions anytime and can click **Consider** to move them to the **Recommended actions** tab.



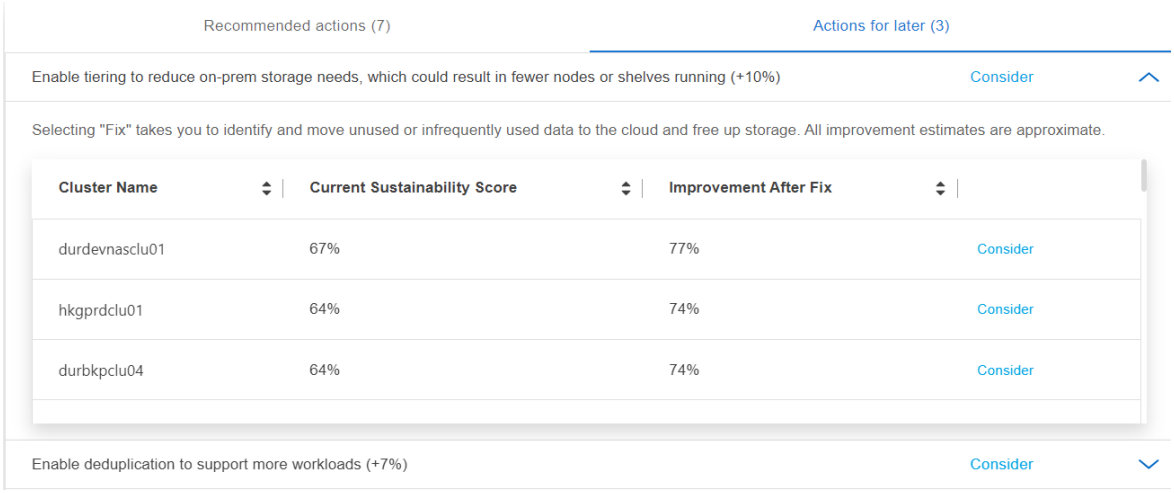
Digital Advisor

1. Go to **Sustainability** from the **STORAGE HEALTH** category available in the left navigation of Digital Advisor dashboard.
2. Go to the **Recommended actions** tab.
   - You can select **Fix** to take these actions immediately, or select **Later** to address these actions later.

- If you plan to address the actions immediately, select **Fix**.
  - The view of the selected recommended action expands. You can also expand the recommended action view using the down arrow. In the expanded view, you can see the cluster name, the sustainability score, and subsequent increase if you proceed with the **Fix** option.



- If you plan to address the actions later, select **Later**.
  - If you select **Later**, it moves the selected recommended action to the **Actions for later** tab. The selected action will be postponed for 30 days. After 30 days, this action will move to the **Recommended actions** tab.
  - You can also review the postponed actions anytime and can click **Consider** to move them to the **Recommended actions** tab.

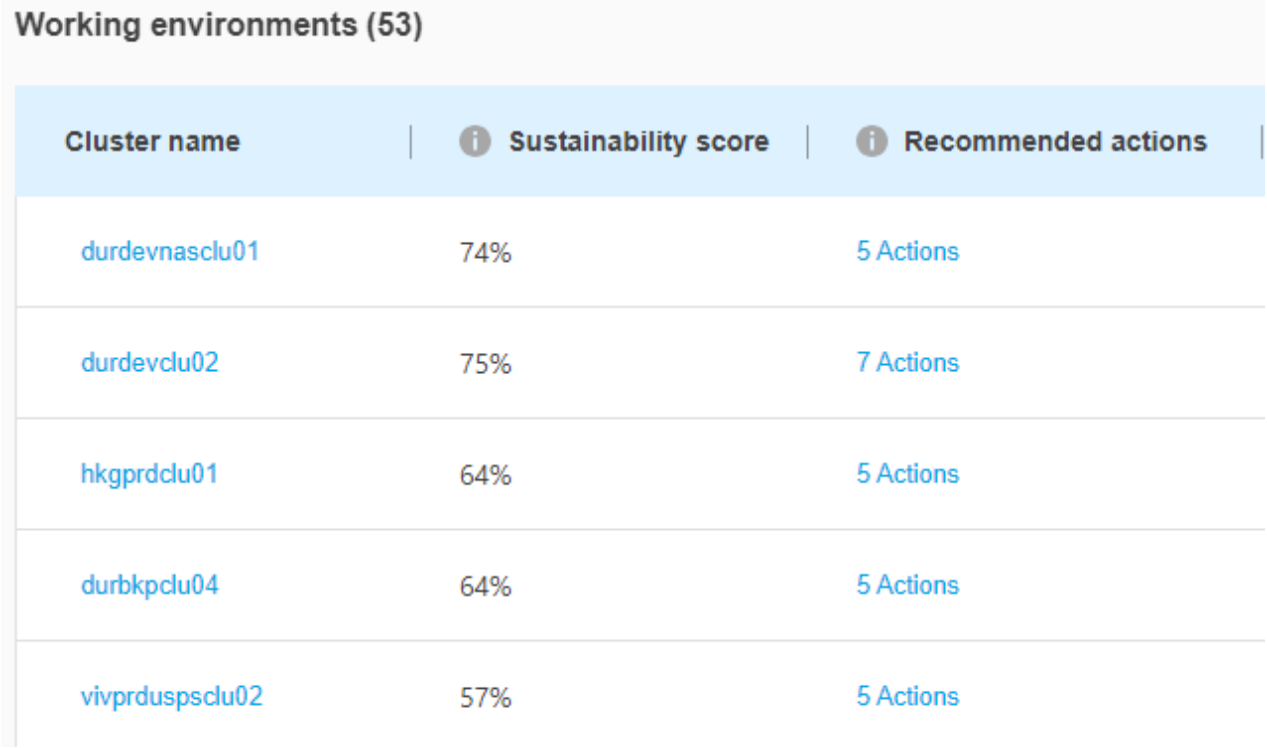


Improve sustainability score at cluster level

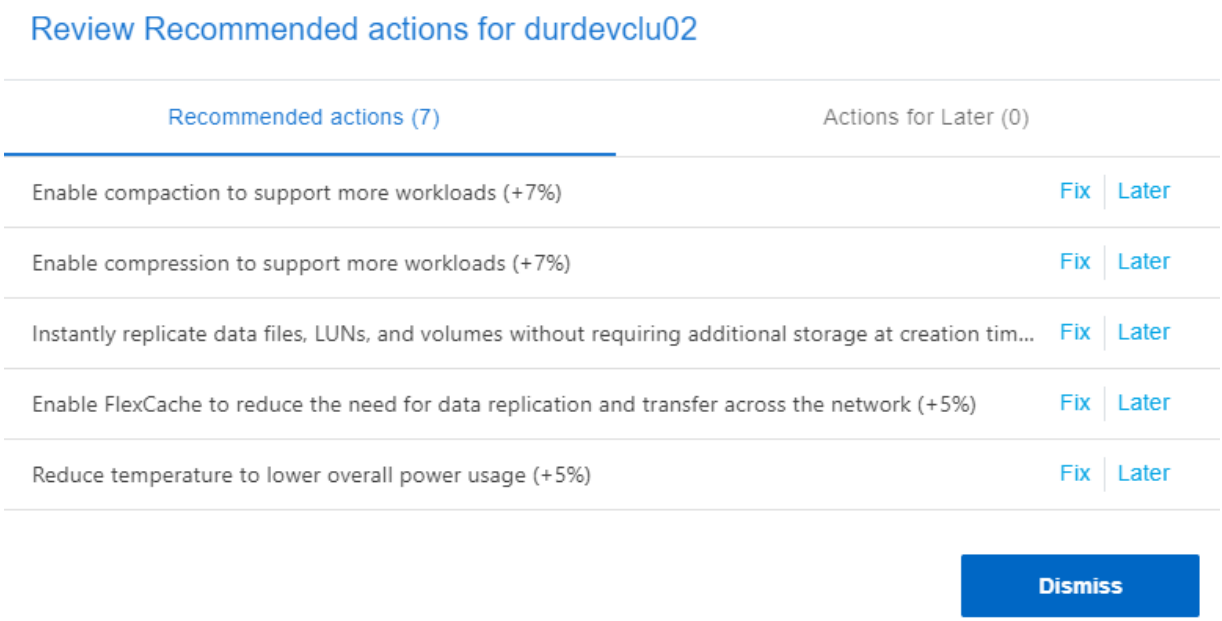
You can follow these steps to improve the cluster-level sustainability score that focuses on improving sustainability for a specific cluster:

BlueXP

1. Go to **Sustainability** from the **Governance** category available in the BlueXP left navigation.
2. Go to the **Working environments** table.
3. Click the number of actions from the **Recommended actions** column for the target cluster.



- You can select **Fix** to take these actions immediately, or select **Later** to address these actions later.



- If you select **Later**, it moves the selected recommended action to the **Actions for later** tab. The selected action will be postponed for 30 days. After 30 days, this action will move to the

Recommended actions tab.

- You can also review the postponed actions anytime and can click **Consider** to move them to the **Recommended actions** tab.

> You can add or remove the environmental parameters in the **Working environments** table using the icon, and can export this table in comma-separated values (.csv) format using the icon.

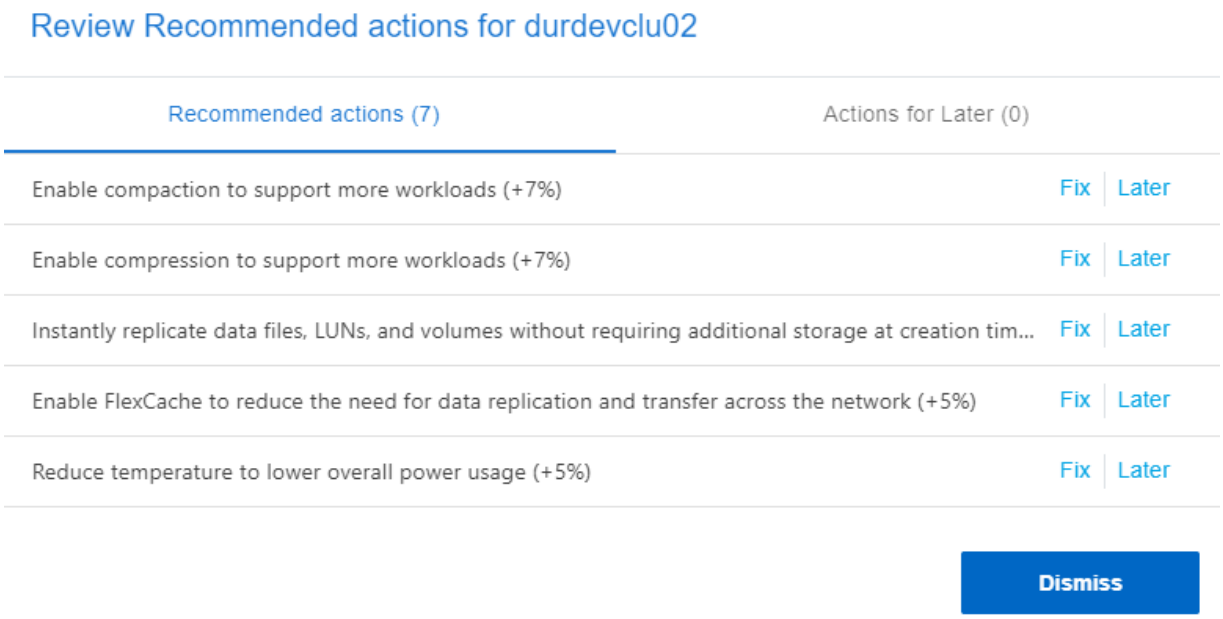
Digital Advisor

1. Go to **Sustainability** from the **STORAGE HEALTH** category available in the left navigation of Digital Advisor dashboard.
2. Go to the **Recommended actions** tab.
3. Go to the **Working environments** table.
4. Click the number of actions from the **Recommended actions** column for the target cluster.

Working environments (53)

Cluster name	Sustainability score	Recommended actions
durdevnasclu01	74%	5 Actions
durdevclu02	75%	7 Actions
hkgprdclu01	64%	5 Actions
durbkpclu04	64%	5 Actions
vivprduspsclu02	57%	5 Actions

- You can select **Fix** to take these actions immediately, or select **Later** to address these actions later.



- If you select **Later**, it moves the selected recommended action to the **Actions for later** tab. The selected action will be postponed for 30 days. After 30 days, this action will move to the **Recommended actions** tab.
- You can also review the postponed actions anytime and can click **Consider** to move them to the **Recommended actions** tab.

> You can add or remove the environmental parameters in the **Working environments** table using the icon, and can export this table in comma-separated values (.csv) format using the icon.

Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.